

# Manual NixOS Installation 101



**DISCLAIMER: THIS GUIDE IS WRITTEN FOR A DUAL-BOOT  
INSTALLATION STARTING ON A WINDOWS OPERATING SYSTEM  
WITH AMD64 ARCHITECTURE. RESULTS MAY VARY ON OTHER  
SYSTEMS.**

# 1. Prerequisite

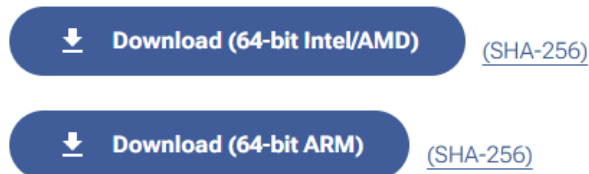
Prior to the installation of NixOS, several prerequisites must be fulfilled.

## 1.1. Bootable USB

First, navigate to the <https://nixos.org/download> page and download a minimal NixOS ISO image. Scroll to the bottom of the page to locate "Minimal ISO Image" and download the ISO compatible with your system architecture; for this guide, the system is based on AMD64.

### Minimal ISO image

The minimal installation ISO image **does not contain the graphical user interface**, and is therefore a lot smaller. You have to [run the installer from the console](#). It contains a number of rescue tools.



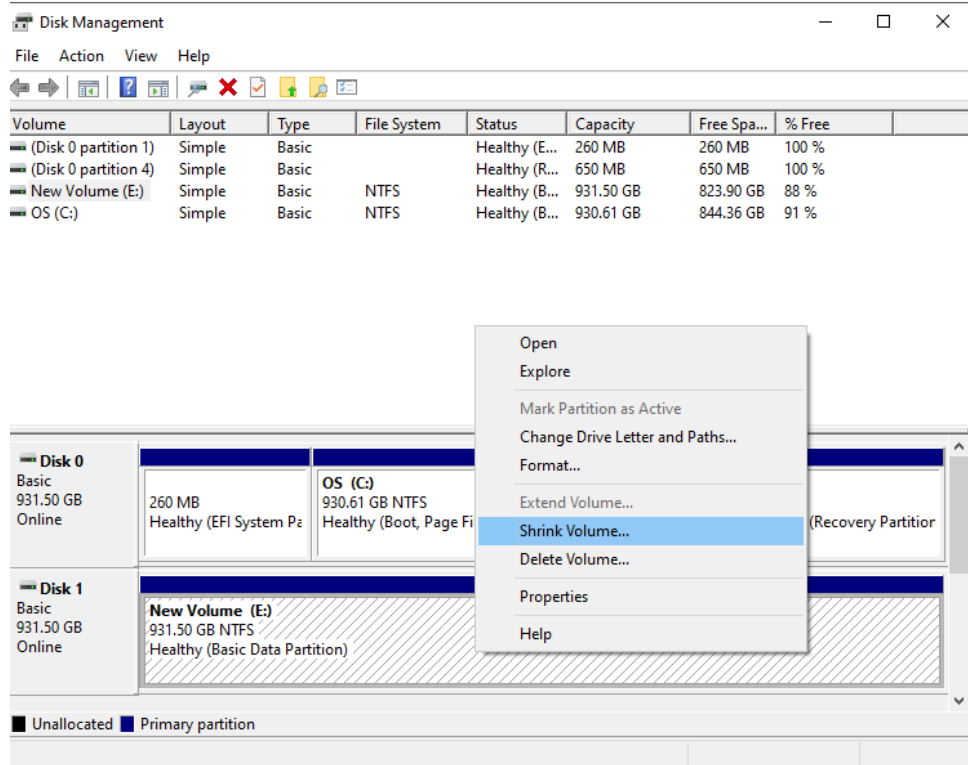
Following the ISO download, utilize Rufus (<https://rufus.ie/en>) to create a bootable USB drive, selecting DD as the formatting option.

Please select the mode that you want to use to write this image:

- ☐ Write in ISO Image mode (Recommended)
- ☒ Write in DD Image mode

## 1.2. Unallocated disk space

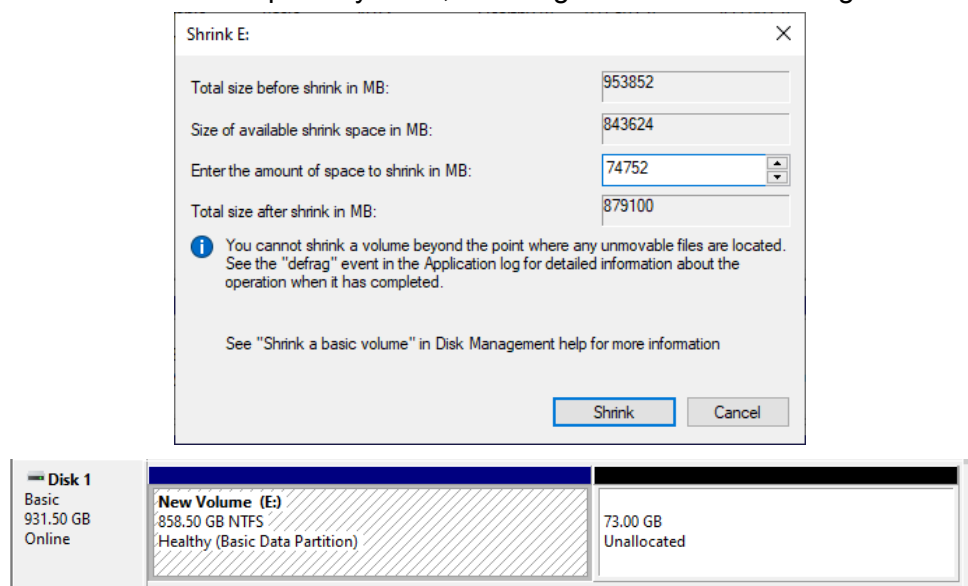
Open "Disk Management" by pressing Windows + R and typing "diskmgmt.msc," then pressing Enter. Within Disk Management, right-click on the target disk and select "shrink volume."



Adjust the allocated space according to your requirements. For the purposes of this guide, the necessary space allocations are as follows:

1. EFI Partition: 1GB (minimum 512MB)
2. SWAP Partition:  $\frac{1}{2}$  the size of RAM (e.g., 8GB for 16GB RAM)
3. Root Partition: 64GB (minimum 32GB to follow this guide, adjustable as needed)

The total required space amounts to 73GB. Since the system calculates in MB, this value must be multiplied by 1024, resulting in 74572MB for this guide.

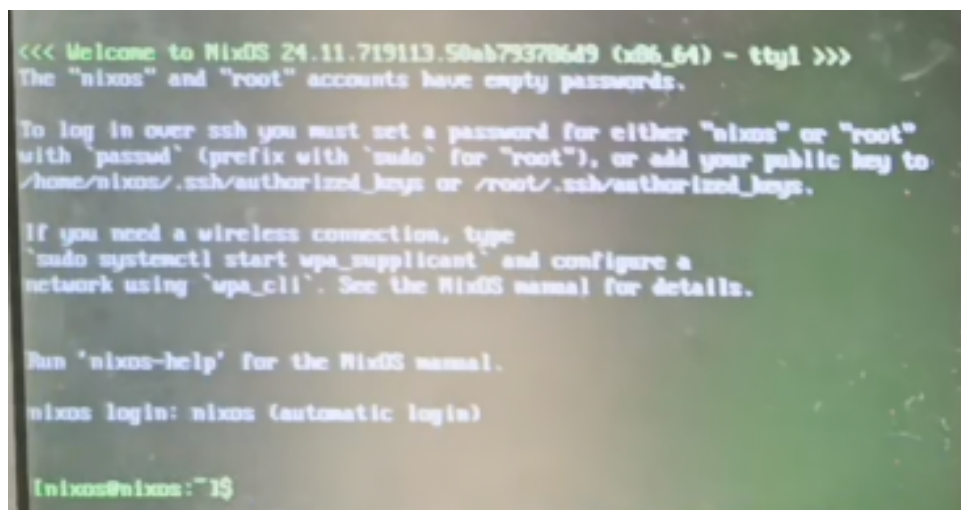
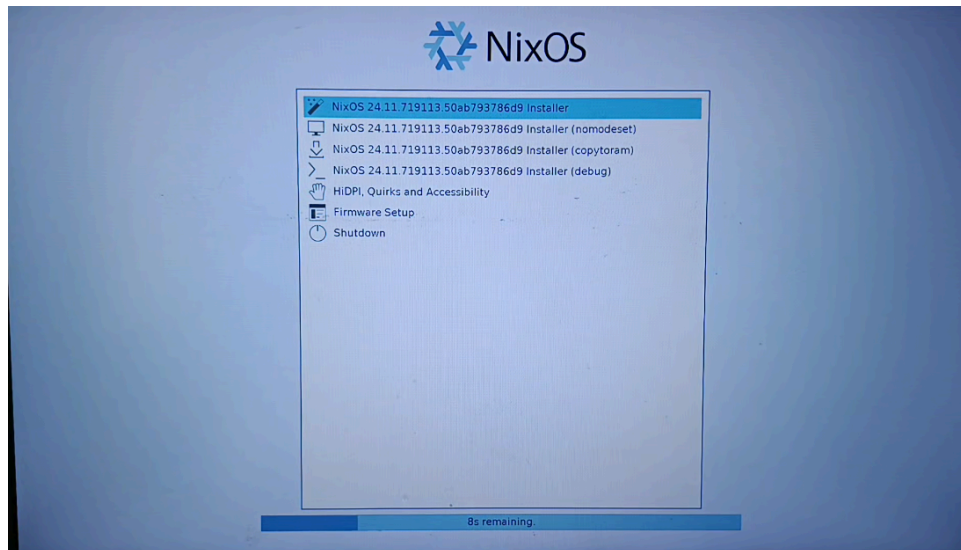


## 2. Installation

### 2.1. NixOS Installation CLI

The very first phase of NixOS installation requires accessing the installation CLI via a live CD/USB ISO image.

To do so, insert your flash drive or any bootable device into your computer and then initiate a reboot. Access the BIOS/UEFI settings (the key may vary depending on your manufacturer, typically F2 or DEL) and disable Secure Boot within the Security or Boot Options. Then, set your boot device as USB Boot.



If you already see things based on the figure above, you may proceed to the next step.



## 2.2. Connecting to Internet

In order to be able to install NixOS, you need access to the internet. You can do this by plugging in an ethernet cable to your device or using wifi. To use a wifi you need to enable it first.

Run **sudo -i** to enter setup environment

```
<<< Welcome to NixOS 24.11.719113.50ab793786d9 (x86_64) - tty1 >>>
The "nixos" and "root" accounts have empty passwords.

To log in over ssh you must set a password for either "nixos" or "root"
with 'passwd' (prefix with 'sudo' for "root"), or add your public key to
/home/nixos/.ssh/authorized_keys or /root/.ssh/authorized_keys.

If you need a wireless connection, type
'sudo systemctl start wpa_supplicant' and configure a
network using 'wpa_cli'. See the NixOS manual for details.

Run 'nixos-help' for the NixOS manual.

nixos login: nixos (automatic login)

[nixos@nixos:~]$ sudo -i
[root@nixos:~]#
```

Then run **ip a** command to list all interfaces in your device

```
[root@nixos:~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eno1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 0c:8c:aa:f7:4c:e9 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
3: wlp4s0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether dc:e9:94:7b:dd:a7 brd ff:ff:ff:ff:ff:ff
```

In this guide the wlan interface is **wlp4s0**. Then run **rfkill unblock all** to unblock all networking interfaces before running **wpa\_supplicant -B -i <your\_wlan\_interface> -c <(wpa\_passphrase 'your\_ssid' 'your\_wifi\_password')>**. If “Successfully initialized wpa\_supplicant” message appeared then you’re connected to the wifi.

```
[root@nixos:~]# rfkill unblock all
[root@nixos:~]# wpa_supplicant -B -i wlp4s0 -c <(wpa_passphrase 'your_ssid' 'your_wifi_password')>
Successfully initialized wpa_supplicant
```

You can check whether you’re already connected to the wifi by running **ip a** again

```

[root@nixos:~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eno1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 0c:0c:aa:c7:4e:e0 brd ff:ff:ff:ff:ff:ff
    altname emp3s0
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 4c:c9:94:7b:44:a7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.135/24 brd 192.168.100.255 scope global dynamic noprefixroute wlan0
        valid_lft 259199sec preferred_lft 226799sec
    inet6 2001:440a:2003:bale:7b2c:80bb:793f:d220/64 scope global temporary dynamic
        valid_lft 259194sec preferred_lft 86309sec
    inet6 2001:440a:2003:bale:dec9:94ff:fc7b:dda7/64 scope global dynamic wngtppaddr noprefixroute
        valid_lft 259194sec preferred_lft 172794sec
    inet6 fe80::dec9:94ff:fc7b:dda7/64 scope link
        valid_lft forever preferred_lft forever

```

Then to check if you have access to internet, simply run **ping google.com** then press ctrl+c to stop the **ping**

```

[root@nixos:~]# ping google.com
PING google.com (2404:6800:4003:c04::64) 56 data bytes
^C
--- google.com ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms

```

## 2.3. Making Partitions

The next step is to make partitions and format it. To do so, run **cgdisk /dev/diskname** to make partitions. In this guide, the partition sizes are 1GB for EFI, 8GB for SWAP, and 64GB for root.

| Sectors   | Size   | Type                         |
|-----------|--------|------------------------------|
| 1083392   | 529M   | Windows recovery environment |
| 204800    | 100M   | EFI System                   |
| 32768     | 16M    | Microsoft reserved           |
| 356352000 | 169.9G | Microsoft basic data         |
| 466006016 | 222.2G | Microsoft basic data         |
| 2097152   | 1G     | EFI System                   |
| 16777216  | 8G     | Linux filesystem             |
| 134217728 | 64G    | Linux filesystem             |

Don't forget to write each partition to alter the partition table. Also check if each partition has been written to the table by running **lsblk**.

```

[root@nixos:~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0   1.2G  1 loop /nix/.ro-store
sda         8:0    1   7.5G  0 disk
├─sda1      8:1    1   1.3G  0 part /iso
├─sda2      8:2    1    3M  0 part
└─sdb       8:16   0 465.0G  0 disk
   ├─sdb1    8:17   0   529M  0 part
   ├─sdb2    8:18   0   100M  0 part
   ├─sdb3    8:19   0    16M  0 part
   ├─sdb4    8:20   0   169.9G  0 part
   ├─sdb5    8:21   0   222.2G  0 part
   ├─sdb6    8:22   0    1G  0 part
   ├─sdb7    8:23   0    8G  0 part
   └─sdb8    8:24   0   64G  0 part

```

Then for each partition run the following command, **mkfs.fat -F 32 /dev/efi\_system\_partition** for EFI partition, **mkswap /dev/swap\_partition** for SWAP partition, and **mkfs.ext4 /dev/root\_partition** for root partition

```
[root@nixos:~]# mkfs.fat -F 32 /dev/sdb6
mkfs.fat 4.2 (2021-01-31)

[root@nixos:~]# mkfs.ext4 /dev/sdb8
mke2fs 1.47.1 (20-May-2024)
/dev/sdb8 contains a ext4 file system
last mounted on / on Sat Aug 9 03:31:55 2025
Proceed anyway? (y,N) y
Creating filesystem with 16777216 4k blocks and 4194304 inodes
Filesystem UUID: 594b342a-00e0-4239-acc3-1cf6e69432b3
Superblock backups stored on blocks:
32768, 98304, 163840, 223376, 294912, 819200, 884736, 1605632, 2654208,
4096000, 7962624, 11239424

Allocating group tables: done
Writing inode tables: done
Creating journal (131072 blocks): done
Writing superblocks and filesystem accounting information: done

[root@nixos:~]# mkswap /dev/sdb7
mkswap: /dev/sdb7: warning: wiping old swap signature.
Setting up swspace version 1, size = 8 GiB (8589930496 bytes)
no label, UUID=917eca44-07e9-4162-aa05-b0f7974ffed5
```

## 2.4. Mount Filesystem

After formatting the partitions the next thing you should do is mount it. For each partitions run the following command, **mount /dev/root\_partition /mnt**, **mkdir -p /mnt/boot/efi** then **mount /dev/efi\_system\_partition /mnt/boot/efi**, and **swapon /dev/swap\_partition**

```
[root@nixos:~]# mount /dev/sdb8 /mnt

[root@nixos:~]# mkdir -p /mnt/boot/efi

[root@nixos:~]# mount /dev/sdb6 /mnt/boot/efi

[root@nixos:~]# swapon /dev/sdb7
```

## 2.5. Generate NixOS Configuration File

This part might be different from any other conventional manual linux distribution installation because of the “immutable” design and “atomic” update model, so be sure to follow it thoroughly.

First, generate the NixOS configuration by running **nixos-generate-config --root /mnt**

```
[root@nixos:~]# nixos-generate-config --root /mnt
writing /mnt/etc/nixos/hardware-configuration.nix...
writing /mnt/etc/nixos/configuration.nix...
For more hardware-specific settings, see https://github.com/NixOS/nixos-hardware.
```

Then edit the configuration file using text editor, in this guide **vim** is used as the text editor

```
[root@nixos:~]# vim /mnt/etc/nixos/configuration.nix
```

(Optional) change this only if you want to use grub as bootloader

```
# Edit this configuration file to define what should be installed on
# your system. Help is available in the configuration.nix(5) man page, on
# https://search.nixos.org/options and in the NixOS manual ('nixos-help').

{ config, lib, pkgs, ... }:

{
  imports =
    [ # Include the results of the hardware scan.
      ./hardware-configuration.nix
    ];

  # Use the systemd-boot EFI boot loader.
  boot.loader.systemd-boot.enable = false;
  boot.loader.efi.canTouchEfiVariables = true;
  boot.loader.efi.efiSysMountPoint = "/boot/efi";
  boot.loader.grub.efiSupport = true;
  boot.loader.grub.device = "vdev";
  boot.loader.grub.useOSProber = true;
}
```

Then you can change your hostname as you like, enable NetworkManager, and set your timezone

```
networking.hostName = "nio"; # Define your hostname.
# Pick only one of the below networking options.
# networking.wireless.enable = true; # Enables wireless support via wpa_supplicant.
networking.networkmanager.enable = true; # Easiest to use and most distros use this by default.

# Set your time zone.
time.timeZone = "Asia/Jakarta";
```

Enable **nix-command** and **flakes**, and ignore any X11 configuration since this guide will be using **Hydenix** as WM



```

# Configure network proxy if necessary
# networking.proxy.default = "http://user:password@proxy:port/";
# networking.proxy.noProxy = "127.0.0.1,localhost,internal.domain";

# Select internationalisation properties.
# i18n.defaultLocale = "en_US.UTF-8";
# console = {
#   font = "Lat2-Terminus16";
#   keymap = "us";
#   useXkbConfig = true; # use xkb.options in tty.
# };

# Enable the X11 windowing system.
# services.xserver.enable = true;

nix.settings.experimental-features = [ "nix-command" "flakes" ];

# Configure keymap in X11
# services.xserver.xkb.layout = "us";
# services.xserver.xkb.options = "eurosign:c,caps:escape";

# Enable CUPS to print documents.
# services.printing.enable = true;

# Enable sound.
# hardware.pulseaudio.enable = true;
# OR
# services.pipewire = {
#   enable = true;
#   pulse.enable = true;
# };

# Enable touchpad support (enabled default in most desktopManager).
# services.libinput.enable = true;

```

Configure this as well as you need a **user account** and as many needed **packages** (see <https://search.nixos.org/packages> to search for available packages) in your system

```

# Define a user account. Don't forget to set a password with #passwd.
users.users.sustuna = {
  isNormalUser = true;
  extraGroups = [ "wheel" ]; # Enable #sudo for the user.
  packages = with pkgs: [
    tree
  ];
};

# programs.firefox.enable = true;

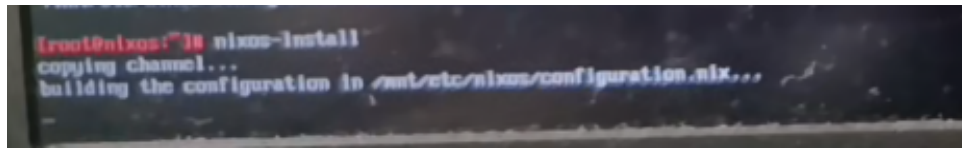
# List packages installed in system profile. To search, run:
# $ nix search wget
environment.systemPackages = with pkgs: [
  vim # Do not forget to add an editor to edit configuration.nix! The
  wget
  git
  ncovim
];

```

Last but not least, write and save your configuration file

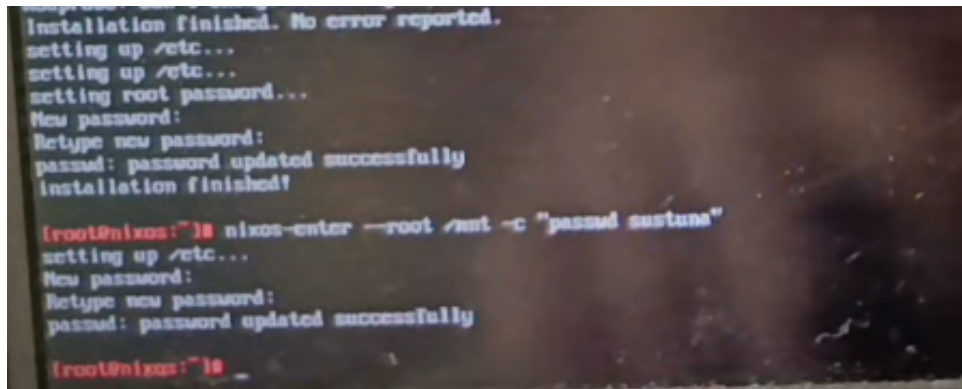
## 2.6. NixOS Install

This is the final step you need to take in order to be able to use a minimal NixOS. To do so all you need is run **nixos-install** and let the magic works



```
[root@nixos:~]# nixos-install
copying channel...
building the configuration in /mnt/etc/nixos/configuration.nix...
```

After the installation finished you need to set your root password, and then set your **user account** password by running **nixos-enter --root /mnt -c "passwd <your\_username>"**

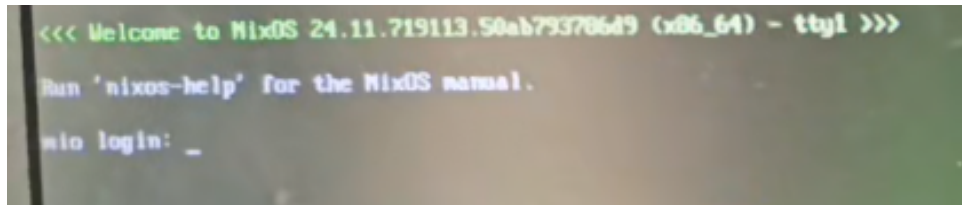


```
Installation finished. No error reported.
setting up /etc...
setting up /etc...
setting root password...
New password:
Retype new password:
passwd: password updated successfully
Installation finished!

[root@nixos:~]# nixos-enter --root /mnt -c "passwd sustuna"
setting up /etc...
New password:
Retype new password:
passwd: password updated successfully

[root@nixos:~]# _
```

Finally, reboot your system by running **reboot** and voila! Your minimal NixOS is ready to use



```
<<< Welcome to NixOS 24.11.719113.50ab793786d9 (x86_64) - tty1 >>>

Run 'nixos-help' for the NixOS manual.

min login: _
```

## 3. Hydenix

To use your NixOS comfortably you will need GUI either it's a Desktop Environment or Windows Manager. In this guide Hydenix will be used as a Windows Manager.

### 3.1. Environment Setup

Before installing Hydenix, you will have to set up the installation environment. First, make a directory for Hydenix configuration files by running **mkdir hydenix && cd hydenix**

```
<<< Welcome to NixOS 24.11.719113.50ab793706d9 (x86_64) - tty1 >>>

Run 'nixos-help' for the NixOS manual.

nix login: sustuna
Password:

[sustuna@nix:~]$ mkdir hydenix && cd hydenix

[sustuna@nix:~/hydenix]$ _
```

Then connect your device to the internet by plugging in an ethernet cable or using wifi. Again, in this guide we will be using wifi.

Run **nmcli** to see the list of your network device names.

```
[sustuna@nix:~/hydenix]$ nmcli
lo: connected (externally) to lo
  "lo"
  loopback (unknown), 00:00:00:00:00:00, sw, mtu 65536
  inet4 127.0.0.1/8
  inet6 ::1/128

wlp1s0: disconnected
  "Realtek RTL8822CE"
  wifi (rtl_8822ce), CA:01:26:0E:BF:D4, hw, mtu 1500

enol: unavailable
  "Realtek RTL8111/8168/8211/8411"
  ethernet (r8169), 0C:0C:AA:77:4E:ED, hw, mtu 1500

Use "nmcli device show" to get complete information about known devices and
"nmcli connection show" to get an overview on active connection profiles.

Consult nmcli(1) and nmcli-examples(7) manual pages for complete usage details.
```

In this guide the device name for the wifi module is **wifi**. Then run **sudo nmcli device <device\_name> connect <ssid> password <password>**

```
[sustuna@nix:~/hydenix]$ sudo nmcli device wifi connect [redacted] password [redacted]

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for sustuna:
Device 'wlp1s0' successfully activated with '452c94da-7e16-4b1f-b53a-5bfc7baa7910'.
```

After connecting to the internet, run **nix flake init -t github:richen604/hydenix**

```

[sustuna@nio:~/hydenix]$ nix flake init -t github:richens04/hydenix
wrote: /home/sustuna/hydenix/.gitignore
wrote: /home/sustuna/hydenix/hardware-configuration.nix
wrote: /home/sustuna/hydenix/README.md
wrote: /home/sustuna/hydenix/flake.nix
wrote: /home/sustuna/hydenix/configuration.nix
wrote: /home/sustuna/hydenix/docs/options.md
wrote: /home/sustuna/hydenix/docs/upgrading.md
wrote: /home/sustuna/hydenix/docs/faq.md
wrote: /home/sustuna/hydenix/docs/assets/option-search.png
wrote: /home/sustuna/hydenix/docs/assets
wrote: /home/sustuna/hydenix/docs/community.md
wrote: /home/sustuna/hydenix/docs/troubleshooting.md
wrote: /home/sustuna/hydenix/docs/installation.md
wrote: /home/sustuna/hydenix/docs/contributing.md
wrote: /home/sustuna/hydenix/docs
wrote: /home/sustuna/hydenix/modules/ha/default.nix
wrote: /home/sustuna/hydenix/modules/ha
wrote: /home/sustuna/hydenix/modules/system/default.nix
wrote: /home/sustuna/hydenix/modules/system
wrote: /home/sustuna/hydenix/modules

```

### 3.2. Configuration File

After initializing the environment, edit the configuration file (configuration.nix) using text editor. In this guide the command is **vim configuration.nix**. Edit these lines based on your system

(Optional) if you're installing NixOS on a laptop or any device that has two GPUs you shall run **nix-shell -p pciutils --run "lspci" | grep VGA** first before editing the config.

```

[sustuna@nio:~/hydenix]$ nix-shell -p pciutils --run "lspci" | grep VGA
01:00.0 VGA compatible controller: NVIDIA Corporation TU117H [GeForce GTX 1650 Ti Mobile] (rev a1)
06:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Renoir [Radeon Vega Series / Radeon Vega Mobile Series] (rev c6)

```

The numbers shown in figure above will be the PCI number for its respective GPU



```

# -----
# GPU Configuration (choose one)
# -----

# NVIDIA GPUs:
# 1. Uncomment the line below:
inputs.hydenix.inputs.nixos-hardware.nixosModules.common-gpu-nvidia

# -----

# AMD GPUs:
# inputs.hydenix.inputs.nixos-hardware.nixosModules.common-gpu-amd

# -----

# CPU Configuration (choose one)
# -----

# AMD CPUs:
inputs.hydenix.inputs.nixos-hardware.nixosModules.common-cpu-amd

# Intel CPUs (also enables Intel graphics):
# inputs.hydenix.inputs.nixos-hardware.nixosModules.common-cpu-intel

# -----

# Additional Hardware Modules
# -----

# Uncomment based on your system type:

# For high-DPI displays (4K, etc.):
# inputs.hydenix.inputs.nixos-hardware.nixosModules.common-hidpi

# For laptops:
inputs.hydenix.inputs.nixos-hardware.nixosModules.common-pc-laptop

# For SSD storage:
inputs.hydenix.inputs.nixos-hardware.nixosModules.common-pc-ssd

# For HDD storage:
inputs.hydenix.inputs.nixos-hardware.nixosModules.common-pc-hdd

# -----

# 2. If you have hybrid graphics (laptop), also set these:
hardware.nvidia.prime.andgpuBusId = "PCI:6:0:0"; # Run: lspci | grep VGA
hardware.nvidia.prime.nvidiaBusId = "PCI:1:0:0"; # to get correct IDs

# -----

# 3. For newer cards, you might want open drivers:
hardware.nvidia.open = true;

```

Then set your **user account**, **hostname**, and **timezone** according to your NixOS Installation and don't forget to set the **localization**

```

# -----
# User Configuration
# -----
# REQUIRED: Change "hydenix" to your actual username
# This must match the username you define in users.users below
users."sustuna" =
{ ... }:
{
  imports = [
    inputs.hydenix.lib.homeModules
    # Command-not-found and corepack tool support
    inputs.nix-index-database.homeModules.nix-index
    ./modules/hm # Your custom home-manager modules
  ];
};

# =====
# USER ACCOUNT SETUP
# =====
# REQUIRED: This creates your user account
# Change "hydenix" to your desired username (must match above)
users.users.sustuna = {
  isNormalUser = true;
  # SECURITY: Change this password after first login with 'passwd'
  initialPassword = "hydenix";

  # User groups (determines permissions):
  extraGroups = [
    "wheel" # Sudo access (admin privileges)
    "networkmanager" # Network configuration
    "video" # Graphics/display access
  ];

  # Default shell (options: pkgx.bash, pkgx.zsh, pkgx.fish)
  shell = pkgx.zsh;
};

# =====
# HYDENIX CONFIGURATION
# =====
# Main configuration for the Hydenix desktop environment
hydenix = {
  enable = true; # Enable Hydenix modules

  # -----
  # Basic System Settings (REQUIRED)
  # -----
  # REQUIRED: Set your computer's network name
  hostname = "nio"; # Change to something unique

  # REQUIRED: Set your timezone
  # Examples: "America/New_York", "Europe/London", "Asia/Tokyo"
  # Full list: `timedatectl list-timezones`
  timezone = "Asia/Jakarta";

  # REQUIRED: Set your locale/language
  # Examples: "en_US.UTF-8", "en_GB.UTF-8", "de_DE.UTF-8"
  locale = "en_US.UTF-8";
};

```

Don't forget to enable **grub** and set **efiSysMountPoint** to **/boot/efi**

```

boot.useSystemdBoot = false;

# -----
# Optional Settings
# -----
# For more configuration options, see:
# https://github.com/richen604/hydenix/blob/main/docs/options.md
);
boot.loader.efi.efiSysMountPoint = "/boot/efi";

```

Then generate hardware configuration file by running **nixos-generate-config --show-hardware-config > hardware-configuration.nix**

```
[sustuna@mio:~/hydenix]$ nixos-generate-config --show-hardware-config > hardware-configuration.nix
```

### 3.3. Hydenix Install

This is the last step on installing Hydenix. Run **sudo nixos-rebuild switch --flake .#hydenix**

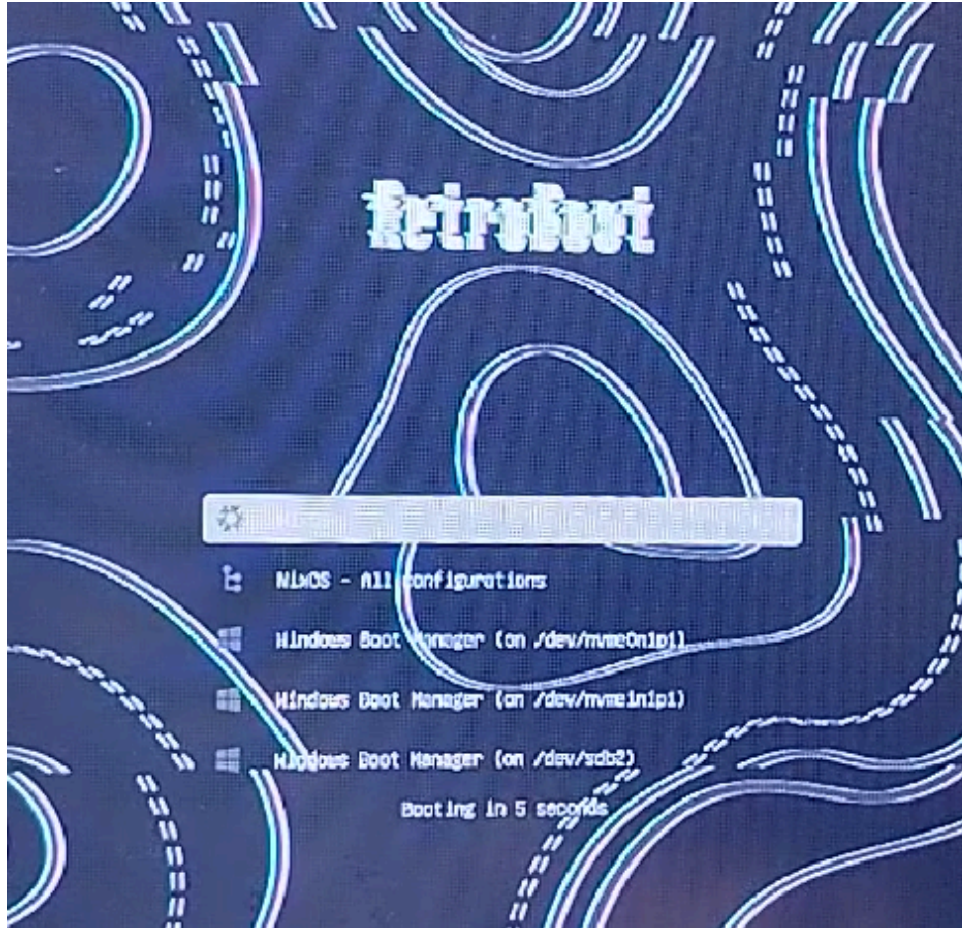
```

[sustuna@mio:~/hydenix]$ sudo nixos-rebuild switch --flake .#hydenix
warning: Git tree '/home/sustuna/hydenix' is dirty
building the system configuration...
warning: Git tree '/home/sustuna/hydenix' is dirty
evaluation warning: nix-index-database: flake output 'modules' has been renamed to 'modules'
evaluation warning: DEPRECATED: The <nixos-hardware/common/bd.nix> module has been deprecated.
This module has no effect and will be removed in a future release.
updating GRUB 2 menu...
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
grub-probe: error: cannot find a GRUB drive for /dev/sda1. Check your device.map.
lsblk: /dev/mapper/no-[0-9]: not a block device
lsblk: /dev/mapper/raid-[0-9]: not a block device
lsblk: /dev/mapper/disk-[0-9]: not a block device
Found Windows Boot Manager on /dev/nvme0n1p18/efi/Microsoft/Boot/bootmgfw.efi
Found Windows Boot Manager on /dev/nvme0n1p18/efi/Microsoft/Boot/bootmgfw.efi
Found Windows Boot Manager on /dev/sdb2/efi/Microsoft/Boot/bootmgfw.efi
stopping the following units: audit.service, kmod-static-nodes.service, logrotate-check
cd.service, resolvconf.service, systemd-modules-load.service, systemd-oomd.service, sys
upa_supplicant.service
NOT restarting the following changed units: getty@tty1.service, systemd-backlight@backli
lush.service, systemd-logind.service, systemd-random-seed.service, systemd-remount-fs.s
activating the configuration...
setting up /etc...
removing obsolete symlink /etc/pass.d/locks...
removing obsolete symlink /etc/pass.d/xlocks...
removing obsolete symlink /etc/pass.d/locks-colors...
removing obsolete symlink /etc/pass.d/xscreensaver...
restarting systemd...
reloading user units for sustuna...
restarting sysinit-reactivation.target
reloading the following units: dbus.service, firewall.service, reload-systemd-console-
restarting the following units: nix-daemon.service, polkit.service, systemd-journald.s
starting the following units: audit.service, kmod-static-nodes.service, logrotate-check
systemd-modules-load.service, systemd-oomd.socket, systemd-sysctl.service, systemd-tim
the following new units were started: acpid.service, bluetooth.service, sshd-unix-local
done. The new configuration is /nix/store/tpch5667b18luqtc9p6ab89q1azrh-nixos-system-

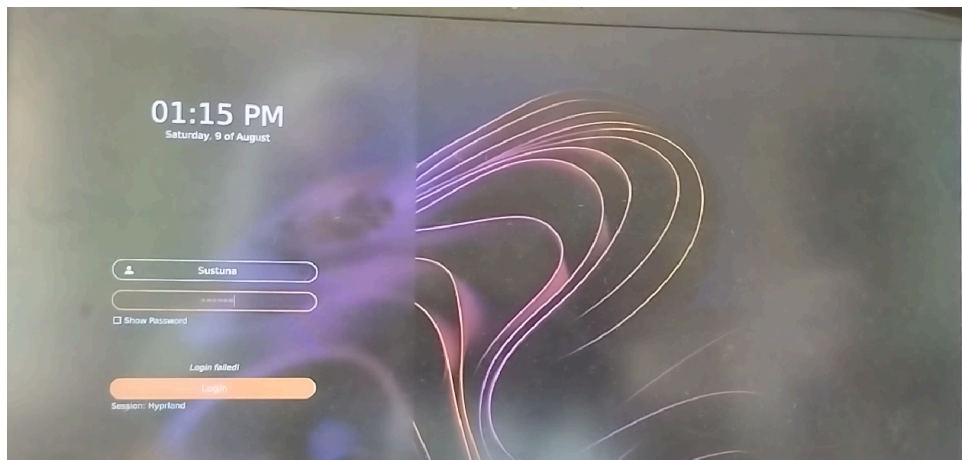
```

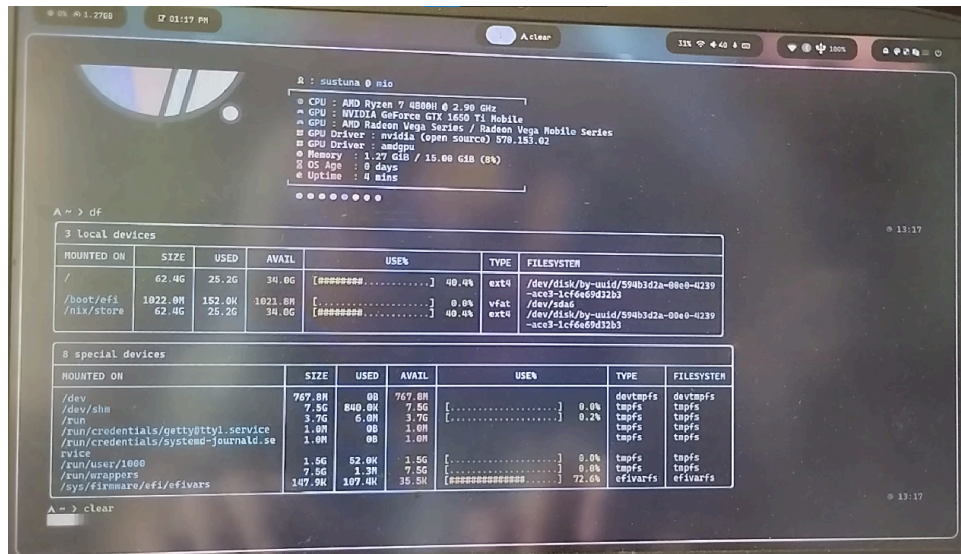
After the installation process finished, run **reboot** to reboot the system





If you see this screen then congrats you've finished installing Hydenix





## References

These are the references used in this guide

1. <https://nixos.org/manual/nixos/stable>
2. <https://github.com/richen604/hydenix/blob/main/template/docs/installation.md>



# ENCORE

## EX1. Dualboot

The guide itself already meant for dualboot by enabling **OSProber** on **GRUB**

## EX2. Changing Wallpaper

Just copy your desired wallpaper to your theme's wallpaper folder

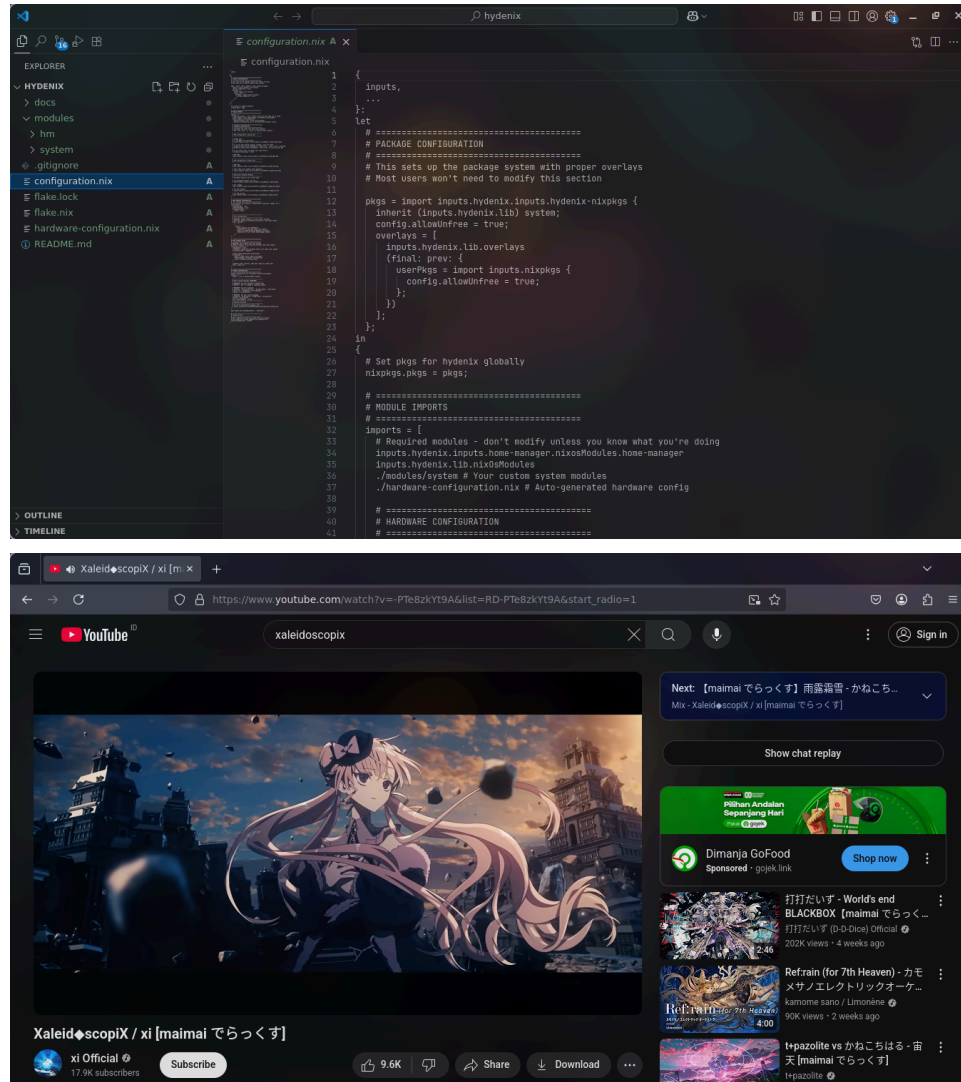
```
A ~/.config > cd hyde
A ~/.config/hyde > ls
config.toml themes wallbash
A ~/.config/hyde > cd themes
A ~/.config/hyde/themes > ls
'Catppuccin Latte' 'Catppuccin Mocha'
A ~/.config/hyde/themes > cd Catppuccin\ Mocha
A hyde/themes/Catppuccin Mocha > ls
hypr.theme Kitty.theme kvantum logo rofi.theme wall.hyperlock.png wallpapers wall.set waybar.theme
A hyde/themes/Catppuccin Mocha > cd wallpapers
A themes/Catppuccin Mocha/wallpapers > ls
'1 rain_world.png' '3 rain_world.png' apex_octane.jpg boat_abandoned.jpg escape_velocity.jpg
'2 rain_world.png' aesthetic_deer.png black_hole_by_kurzgesagt.png cat_leaves.png evening_sky.png
A themes/Catppuccin Mocha/wallpapers > cp ~/Downloads/1310319.png ./
A themes/Catppuccin Mocha/wallpapers > ls
1310319.png '2 rain_world.png' aesthetic_deer.png black_hole_by_kurzgesagt.png cat_leaves.png
'1 rain_world.png' '3 rain_world.png' apex_octane.jpg boat_abandoned.jpg escape_velocity.jpg
A themes/Catppuccin Mocha/wallpapers > |
```

Then press **mod key + shift + w** to change wallpaper



## EX3. Graphical Text Editor and Browser

Hydenix already offered VSCode and Firefox



## EX4. Bootloader Customization

This guide uses Yuuka from BlueArchive as its theme (<https://github.com/ylhcqN/BlueArchive-Grub-theme-yuuka>). Run `nurl https://github.com/ylhcqN/BlueArchive-Grub-theme-yuuka` to obtain owner, repo, rev, and hash information

```
^ ~ > nurl https://github.com/ylhcqN/BlueArchive-Grub-theme-yuuka
$ nix flake prefetch --extra-experimental-features 'nix-command flakes' --json q1
fetchFromGitHub {
  owner = "ylhcqN";
  repo = "BlueArchive-Grub-theme-yuuka";
  rev = "0774651d3070d2f32409a1ef2f1240209140b451";
  hash = "sha256-YMsSZbyP+IptYH9W9aTh4dmnkj3UtM1bM09RZN5N7zY=";
}
```

Then edit **configuration.nix** file in **hydenix** folder

```
locale = "en_US.UTF-8";
boot.enable = false;
# -----
# Optional Settings
# -----
# For more configuration options, see:
# https://github.com/richen604/hydenix/blob/main/docs/options.md
};

boot.loader.systemd-boot.enable = false;
boot.loader.efi.canTouchEfiVariables = true;
boot.loader.efi.efiSysMountPoint = "/boot/efi";
boot.loader.grub.efiSupport = true;
boot.loader.grub.device = "nodev";
boot.loader.grub.useOSProber = true;
boot.loader.grub.theme = pkgs.stdenv.mkDerivation {
  pname = "yuuka-en";
  version = "1.0";
  src = pkgs.fetchFromGitHub {
    owner = "ylhcqN";
    repo = "BlueArchive-Grub-theme-yuuka";
    rev = "0774651d3070d2f32409a1ef2f1240209140b451";
    hash = "sha256-YMsSZbyP+IptYH9W9aTh4dmnkj3UtM1bM09RZN5N7zY=";
  };
  installPhase = "cp -r ./yuuka-en/ $out";
};
```

Run **sudo nixos-rebuild switch --flake .#hydenix** to apply the theme

```
^ ~/hydenix  master * + > sudo nixos-rebuild switch --flake .#hydenix
[sudo] password for sustuna:
building the system configuration...
warning: Git tree '/home/sustuna/hydenix' is dirty
evaluation warning: nix-index-database: flake output 'hmModules' has been renamed to 'homeModules'
evaluation warning: DEPRECATED: The <nixos-hardware/common/hdd.nix> module has been deprecated.

This module has no effect and will be removed in a future release.

updating GRUB 2 menu...
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
lsblk: /dev/mapper/no*[0-9]: not a block device
lsblk: /dev/mapper/raid*[0-9]: not a block device
lsblk: /dev/mapper/disks*[0-9]: not a block device
Found Windows Boot Manager on /dev/sda1/efi/Microsoft/Boot/bootmgfw.efi
activating the configuration...
setting up /etc...
reloading user units for sustuna...
restarting sysinit-reactivation.target
the following new units were started: sysinit-reactivation.target, systemd-tmpfiles-resetup.service
Done. The new configuration is /nix/store/mn0yx64nd1xbj8zqmi2pz98ans11222m-nixos-system-mio-25.11.20250617.c539ae8
```

Voila! Yuuka is now the GRUB Theme

## EX5. Wine + Line

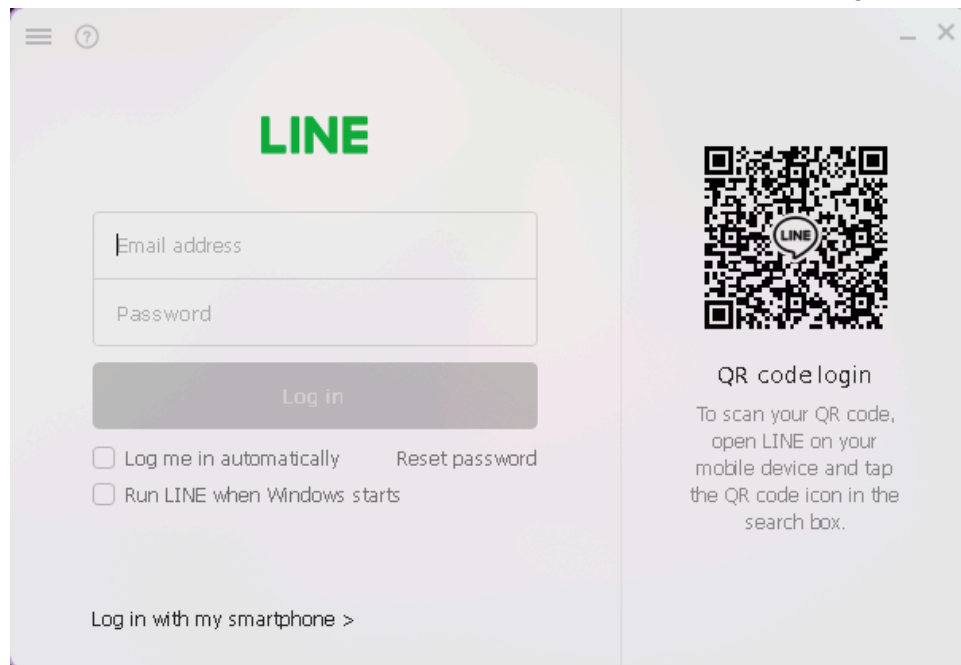
Add `wineWowPackages.stable` and `winetricks` into `hydenix/hydenix/system/default.nix` then rebuild using `sudo nixos-rebuild switch --flake .#hydenix`

```
default.nix
{ pkgs, ... }:

{
  imports = [
    # ./example.nix - add your modules here
  ];

  environment.systemPackages = with pkgs; [
    # pkgs.vscode - hydenix's vscode version
    # pkgs.userPkgs.vscode - your personal nixpkgs version
    nix
    wineWowPackages.stable
    winetricks
  ];
}
```

Download Line Installer from <https://line.en.softonic.com/> and install using **Wine**



## EX6. Tiling Window Manager

Hyprland itself is already a tiling WM



## EX7. Using Wayland

Hyprrland itself is Wayland

## EX8. Watching Anime from CLI

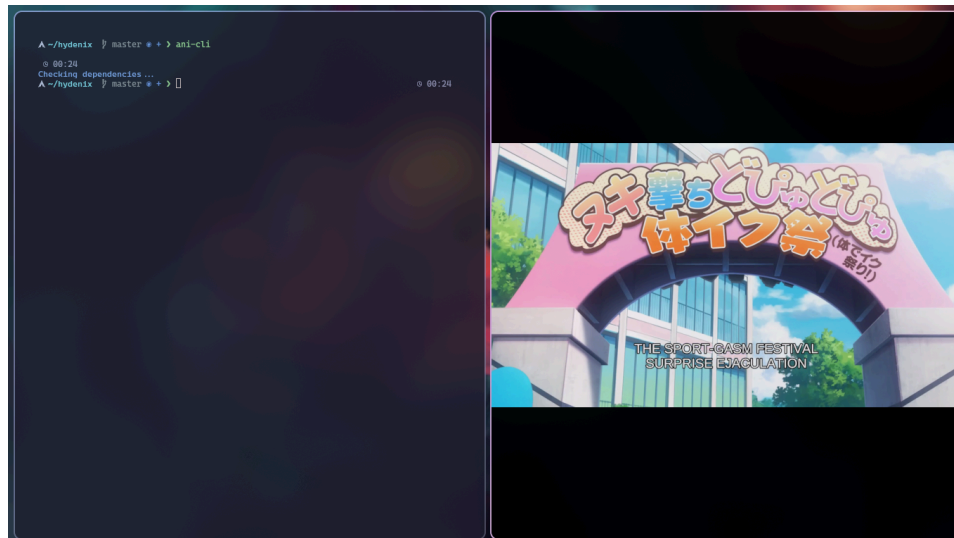
Add **ani-cli** into **hydenix/hydenix/system/default.nix** then rebuild using **sudo nixos-rebuild switch --flake .#hydenix**

```
default.nix
{ pkgs, ... }:

{
  imports = [
    # ./example.nix - add your modules here
  ];

  environment.systemPackages = with pkgs; [
    # pkgs.vscode - hydenix's vscode version
    # pkgs.userPkgs.vscode - your personal nixpkgs version
    nurl
    wineWowPackages.stable
    winetricks
    ani-cli
  ];
}
```

And now enjoy watching anime from CLI



## EX9. Installing and Playing DOOM in Terminal

Add **zig** into **hydenix/hydenix/system/default.nix** then rebuild using **sudo nixos-rebuild switch --flake .#hydenix**

```
{ pkgs, ... }:

{
  imports = [
    # ./example.nix - add your modules here
  ];

  environment.systemPackages = with pkgs; [
    # pkgs.vscode - hydenix's vscode version
    # pkgs.userPkgs.vscode - your personal nixpkgs version
    nurl
    wineWowPackages.stable
    winetricks
    ani-cli
    zig
  ];
}
```

Clone this repository <https://github.com/cryptocode/terminal-doom> and follow the instructions to **build** and **run**

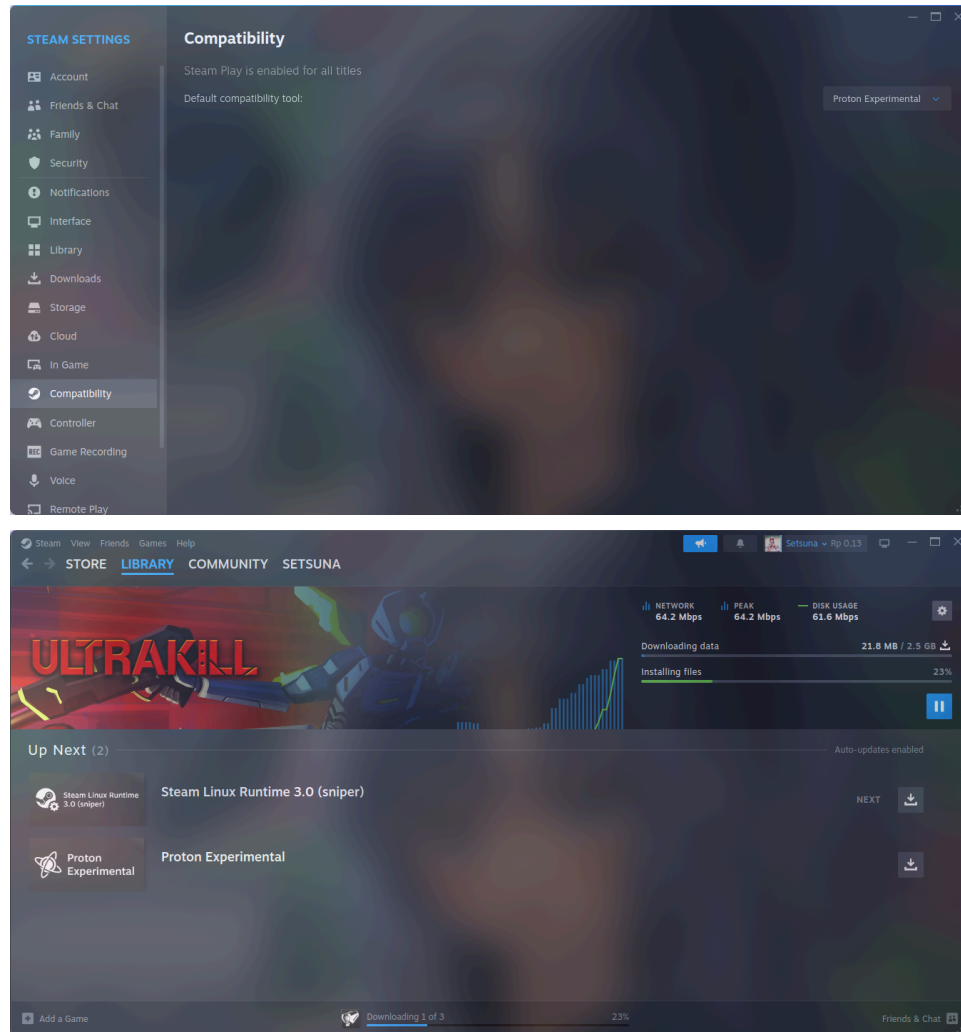
```
A ~ > git clone https://github.com/cryptocode/terminal-doom
Cloning into 'terminal-doom' ...
remote: Enumerating objects: 378, done.
remote: Counting objects: 100% (77/77), done.
remote: Compressing objects: 100% (59/59), done.
remote: Total 378 (delta 50), reused 36 (delta 18), pack-reused 301 (from 1)
Receiving objects: 100% (378/378), 13.87 MiB | 10.82 MiB/s, done.
Resolving deltas: 100% (80/80), done.
A ~ > cd terminal-doom
A ~/terminal-doom ʘ main > █
```

```
A ~/terminal-doom ʘ main > zig build -Doptimize=ReleaseFast
[5] Compile Build Script
├─ [6817] Semantic Analysis
│   └─ multi_array_list.MultiArrayList(array_ha
│       └─ multi_array_list.MultiArrayList(array_ha
└─ [1650] Code Generation
```



## EX10. Installing and Playing Non-Native Linux Game

Hydenix already offered **Steam**, in this guide **ULTRAKILL** will be used as **Non-Native Linux Game**



To play non-native Linux games, set steam **Compatibility** settings to use **Proton** and download **ULTRAKILL**





## EX11. Repacking Installed Linux Distro into an ISO

To repack an installed NixOS into an ISO, you need to modify **flake.nix** file in **hydenix** directory to include installed NixOS configurations and a minimal installation ISO

```
> hydenix > ⚙ flake.nix
1  {
2    description = "A NixOS flake with Hydenix for building a custom installation ISO";
3
4    inputs = {
5      nixpkgs.url = "github:nixos/nixpkgs/nixos-unstable";
6      hydenix = [
7        # Available inputs:
8        # Main: github:richen604/hydenix
9        # Dev: github:richen604/hydenix/dev
10       # Commit: github:richen604/hydenix/<commit-hash>
11       # Version: github:richen604/hydenix/v1.0.0
12       url = "github:richen604/hydenix";
13     ];
14     nix-index-database = {
15       url = "github:nix-community/nix-index-database";
16       inputs.nixpkgs.follows = "nixpkgs";
17     };
18     home-manager = {
19       url = "github:nix-community/home-manager";
20       inputs.nixpkgs.follows = "nixpkgs";
21     };
22   };
23
24   outputs =
25     { ... }@inputs:
26     let
27       hydenixConfig = inputs.hydenix.inputs.hydenix-nixpkgs.lib.nixosSystem {
28         inherit (inputs.hydenix.lib) system;
29         specialArgs = {
30           inherit inputs;
31         };
32         modules = [
33           ./configuration.nix
34           inputs.home-manager.nixosModules.home-manager
35         ];
36       };
37
```

```

38     isoConfig = inputs.hydenix.inputs.hydenix-nixpkgs.lib.nixosSystem {
39         inherit (inputs.hydenix.lib) system;
40         specialArgs = {
41             inherit inputs;
42         };
43         modules = [
44             "${inputs.hydenix.inputs.hydenix-nixpkgs}/nixos/modules/installer/cd-dvd/installation-cd-minimal.nix"
45
46             ./configuration.nix
47             inputs.hydenix.nixosModules.default
48
49             { networking.wireless.enable = false; }
50         ];
51     };
52 };
53 in
54 {
55     nixosConfigurations.hydenix = hydenixConfig;
56
57     nixosConfigurations.iso = isoConfig;
58 };
59 }

```

Then, simply run **sudo nix build** **.#nixosConfigurations.iso.config.system.build.isoImage** and wait until the ISO is being built, the ISO output located at **result/iso**

```

A ~/hydenix  master ④ 17:02 > git add .
A ~/hydenix  master ④ 17:02 > sudo nix build .#nixosConfigurations.iso.config.system.build.isoImage
[sudo] password for sustuna:
warning: Git tree '/home/sustuna/hydenix' is dirty
evaluation warning: nix-index-database: flake output 'hmModules' has been renamed to 'homeModules'
evaluation warning: DEPRECATED: The <nixos-hardware/common/hdd.nix> module has been deprecated.

This module has no effect and will be removed in a future release.
A ~/hydenix  master ④ 17:02 > ls
backup.nix  configuration.nix  docs  flake.lock  flake.nix  hardware-configuration.nix  modules  README.md  result
A ~/hydenix  master ④ 17:02 > cd result
A ~/hydenix/result  > ls
iso  nix-support
A ~/hydenix/result  > cd iso
A hydenix/result/iso  > ls
nixos-minimal-25.11.20250617.c539ae8-x86_64-linux.iso
A hydenix/result/iso  >

```

## Checklist Table

| Specs                                         | Completion                          |
|-----------------------------------------------|-------------------------------------|
| GUI (Mandatory)                               | <input checked="" type="checkbox"/> |
| Audio-Visual (Mandatory)                      | <input checked="" type="checkbox"/> |
| Internet Access (Mandatory)                   | <input checked="" type="checkbox"/> |
| Package Management (Mandatory)                | <input checked="" type="checkbox"/> |
| Unprivileged User (Mandatory)                 | <input checked="" type="checkbox"/> |
| Graphical Text Editor (Mandatory)             | <input checked="" type="checkbox"/> |
| Graphical Web Browser (Mandatory)             | <input checked="" type="checkbox"/> |
| Custom Wallpaper (Mandatory)                  | <input checked="" type="checkbox"/> |
| Instagram Post (Bonus)                        | <input checked="" type="checkbox"/> |
| Youtube Video (Bonus)                         | <input checked="" type="checkbox"/> |
| Graphical Text Editor and Web Browser (Bonus) | <input checked="" type="checkbox"/> |
| Bootloader Customization (Bonus)              | <input checked="" type="checkbox"/> |
| Wine + Line (Bonus)                           | <input checked="" type="checkbox"/> |
| Tiling WM (Bonus)                             | <input checked="" type="checkbox"/> |
| Wayland (Bonus)                               | <input checked="" type="checkbox"/> |
| Watching Anime From CLI (Bonus)               | <input checked="" type="checkbox"/> |
| Playing DOOM in Terminal (Bonus)              | <input checked="" type="checkbox"/> |
| Playing Non-Native Linux Game (Bonus)         | <input checked="" type="checkbox"/> |
| Repacking Installation into an ISO (Bonus)    | <input checked="" type="checkbox"/> |
| Installing in Hardware (Bonus)                | <input checked="" type="checkbox"/> |
| Creativity (Bonus)                            | <input checked="" type="checkbox"/> |

## Links

Youtube Video : <https://youtu.be/pLIGMR8Gnik>

Instagram Post :

<https://www.instagram.com/p/DNTSEZ6TTj7/?igsh=NzRlemNzOWp3azVs>